

HOW NON-RUMINANTS DIFFER — THE GOVERNING CONTRAST

POINT	RUMINANT	NON-RUMINANT
Fibre digestion	Excellent — microbial cellulase	Poor to nil (horse & rabbit are partial exceptions — hind-gut fermenters)
Protein quality	Levelled by microbial protein; degradability matters	Amino-acid balance is critical — lysine, methionine, threonine, tryptophan
NPN / urea	Usable	Toxic — never feed
B vitamins & Vit K	Synthesised in the rumen (needs Co)	Must be supplied in the diet
Energy expressed as	TDN / ME / NE _L	ME (kcal/kg) for poultry & pigs; DE for horses
Phytate phosphorus	Released by rumen phytase	Unavailable ⇒ add phytase or inorganic P

The **ideal protein concept** — the modern basis of pig and poultry formulation: express every essential amino acid as a **percentage of lysine** (lysine = 100), so the diet contains amino acids in exactly the proportion the animal deposits, with no excess to be deaminated. Typical ideal pattern for the pig: **Lys 100 : Met+Cys 60 : Thr 65 : Trp 18**. Formulate on **digestible (ileal) amino acids**, not total.

METHODS OF DETERMINING NUTRIENT REQUIREMENTS

METHOD	PRINCIPLE & COMMENT
Factorial	Requirement is built up by adding the separate needs for maintenance, growth, reproduction and production, each divided by its efficiency of utilisation. Logical and flexible; depends on accurate efficiency values
Dose-response / graded levels	Groups are fed graded levels of the nutrient; performance is plotted and the break-point (plateau) read off. The commonest method for amino acids, vitamins and minerals in poultry
Nitrogen (or mineral) balance	The intake giving zero balance is the maintenance requirement; the level giving maximum retention is the production requirement
Comparative & serial slaughter	Body composition of slaughtered groups reveals actual nutrient deposition . Accurate but destructive and costly
Tissue saturation / enzyme & blood assay	The intake at which the tissue level or the dependent enzyme activity plateaus (e.g. plasma vitamin, glutathione peroxidase for Se)
Radioisotope dilution	Labelled nutrient traces turnover, pool size and true absorption

The **distinction to state clearly**: a **requirement** is the minimum that prevents deficiency and supports the defined performance; a **recommended allowance** is the requirement **plus a margin of safety** for stress, variation in feed composition, bioavailability and processing losses. NRC publishes requirements; BIS and commercial standards publish allowances.

INDIAN STANDARDS — BIS & ICAR SPECIFICATIONS YOU CAN QUOTE

FEED	CLASS	CP (MIN)	ENERGY	OTHER LIMITS
Compounded cattle feed BIS IS:2052	Type I	22%	TDN 70%	Crude fibre max 12% · Acid insoluble ash max 4% · Salt max 2% · Moisture max 11%
	Type II	20%	TDN 65%	
Broiler feed BIS IS:1374	Starter (0–3 wk)	23%	ME 2800–3000 kcal/kg	Ca 1.0% · Available P 0.45% · Lysine 1.1–1.2% · Methionine 0.45–0.50%
	Finisher (4–6 wk)	20%	ME 3000–3200 kcal/kg	
Layer feed BIS IS:1374	Chick mash (0–8 wk)	20%	ME ~2600	Ca 1.0%
	Grower mash (9–20 wk)	16%	ME ~2500	Ca 1.0%
	Layer mash (>20 wk)	18%	ME 2600–2700	Ca 3.5–4.0% · Available P 0.35%

Calorie : protein ratio (kcal ME per kg ÷ % CP) — broiler starter ≈ **130–140**, broiler finisher ≈ **150–160**, layer ≈ **150–170**. **Birds eat to satisfy their energy requirement**: if the ratio widens (energy too high relative to protein) the bird stops eating before it has taken in enough protein and becomes fat with poor growth or small eggs; if it narrows, expensive protein is burnt as energy. **Always change protein and energy together**.

SWINE — THE EFFICIENT OMNIVORE

Digestive peculiarities: simple stomach, omnivorous, **highly efficient converter** (FCR 2.5–3.5), but almost no ability to use fibre (keep CF below 6–8% in young pigs). **Lysine is the first limiting amino acid** in cereal-based diets. Prolific and fast-growing ⇒ nutrition dominates profitability.

CLASS	CP	ME KCAL/KG	FEEDING NOTE
Creep (piglet, 1–3 wk –wean)	20–22%	3300	Offer from 7–10 days ; highly palatable, milk-based; teaches the piglet to eat dry feed
Starter (wean–20 kg)	18–20%	3200	Weaning at 4–8 weeks ; post-weaning check & scours are the main risks
Grower (20–50 kg)	16–18%	3100	<i>Ad libitum</i> ; lean tissue growth phase
Finisher (50–90 kg)	14–16%	3100	Slight restriction to prevent excess back-fat ; withdraw additives before slaughter
Pregnant sow	14%	3000	Restricted — about 2 kg/day . Over-feeding causes dystocia, crushed piglets, low milk and poor rebreeding
Lactating sow	16–18%	3200	<i>Ad libitum</i> — roughly 2 kg + 0.25 kg per suckling piglet ; the highest-demand class on the farm
Breeding boar	14–16%	3000	2–2.5 kg/day ; keep in breeding condition, never fat

Piglet must-knows: ① **Iron-dextran 150–200 mg IM on day 3** — sow's milk is iron-poor and confined piglets have no soil contact ⇒ otherwise **piglet anaemia**. ② Colostrum within 12 h. ③ Creep feed early. ④ Clean water always. **Flushing** the gilt/sow (extra feed 10–14 days before mating) raises ovulation rate.

Deficiency signs unique to pigs: **parakeratosis** (Zn, aggravated by high dietary Ca), **goose-stepping** (pantothenic acid), **mulberry heart & hepatitis dietetica** (Vitamin E/Se), **salt poisoning** (water deprivation with normal salt intake), and **gossypol toxicity** from cottonseed cake.

EQUINE — THE HIND-GUT FERMENTER

Digestive peculiarities — the six that explain every feeding rule: ① **Small stomach** (8–15 L, only ~10% of GIT capacity) ⇒ small frequent meals. ② **No gall bladder** ⇒ continuous bile flow, poor tolerance of large fat loads. ③ Enormous **caecum and colon** (~60% of GIT) housing the microbial population ⇒ fibre is fermented **after** the small intestine, so microbial protein and B vitamins are synthesised **too far down to be well used**. ④ Therefore **dietary protein quality matters** (lysine is first limiting) unlike in a ruminant. ⑤ Cannot vomit. ⑥ Grazing animal designed to eat for 16 h a day.

CLASS	CP	FEEDING NOTE
Foal / creep	16–18%	Creep from 1–2 months; needs high-quality protein & lysine, Ca and P for skeletal growth
Weanling / Yearling	14–16%	Steady growth — over-feeding energy causes developmental orthopaedic disease
Broodmare	12–14%	Extra in the last 3 months of gestation and heavily in early lactation (mare milk peaks at 2 months)
Stallion	12–14%	Maintenance +25% in the breeding season; keep fit, not fat
Working / race horse	10–12%	Energy-dense with fat and grain; electrolytes (Na, K, Cl) after sweating ; protein need rises only slightly

The five golden rules of horse feeding: ① **Feed little and often** (3–4 meals). ② **Roughage first and always — a minimum of 1% of body weight as long-stem forage**, ideally 1.5–2%. ③ **Make every change gradually** over 7–14 days. ④ **Water before feeding**, and never work a horse immediately after a full feed. ⑤ Feed **by weight, not by volume**, and keep to a fixed routine. Total DM intake **1.5–3% of body weight**.

Nutritional disorders of the horse: **Colic** (sudden change, mouldy feed, sand, inadequate water) · **Laminitis** (grain overload, lush pasture — hind-gut acidosis) · **Azoturia** / **"Monday morning disease"** / **tying-up** (full grain ration continued during a rest day, then sudden work) · **"Big head"** (**osteodystrophia fibrosa**) from **high-oxalate** tropical grasses binding calcium · **Monensin poisoning** — an ionophore intended for cattle is **fatal** to horses.

BROILER FEEDING

Peculiarities of the fowl: no teeth (the gizzard grinds), very rapid passage (~4 h), **no urinary separation** — urine and faeces void together as uric acid, so a true urinary N balance is impossible; **ME is used, not TDN**; requires **glycine/serine** in addition to the ten essentials; needs **Vitamin D₃** (D₂ is only ~1/10 as potent); synthesises its own Vitamin C.

PHASE	AGE	CP	ME KCAL/KG
Pre-starter	0–7 d	23–24%	2900–3000
Starter	0–3 wk	23%	2800–3000
Finisher	4–6 wk	20%	3000–3200

Performance benchmarks: market weight **1.8–2.2 kg at 5–6 weeks**; **FCR 1.5–1.7**; mortality <5%. Feed is offered *ad libitum* as **crumbles then pellets**.

Key nutritional points: methionine is the **first limiting amino acid** in maize–soya diets, lysine second; energy is raised through the phases with added fat/oil; **coccidiostat** is included and must be **withdrawn before slaughter**; Ca 1.0% and available P 0.45%; phytase is routinely added to release phytate P.

Ascites (waterbelly) and **Sudden Death Syndrome** occur because modern broilers grow faster than their cardiopulmonary capacity — controlled by moderating early energy intake and lighting. **Skeletal problems (tibial dyschondroplasia, rickets)** follow Ca:P imbalance or Vitamin D₃ deficiency.

LAYER FEEDING

PHASE	AGE	CP	ME	CA
Chick mash	0–8 wk	20%	2600	1.0%
Grower mash	9–20 wk	16%	2500	1.0%
Layer mash	>20 wk	18%	2600–2700	3.5–4.0%

Intake & output: a layer eats **110–120 g/day** and lays an egg whose shell contains about **2 g of calcium** — roughly **10% of her entire body calcium every single day**. That figure alone explains the whole calcium strategy.

Calcium strategy — the marks-fetching detail: supply part of the calcium as **coarse particulate limestone or oyster shell** rather than all as fine powder. Coarse particles are retained in the gizzard and dissolve slowly, releasing calcium **during the night** when shell calcification actually occurs. Feed the calcium-rich portion in the **afternoon/evening**. Also ensure Vitamin D₃, phosphorus and manganese.

Grower restriction: pullets are restricted in the growing phase to prevent over-fat birds, delay sexual maturity to the target age and achieve uniform body weight at housing — uniformity, not weight, predicts peak production.

Layer disorders to name: **Cage layer fatigue** (Ca/P/D₃ — osteoporosis and paralysis in caged birds), **thin/soft shells** (Ca, D₃, heat stress), **fatty liver haemorrhagic syndrome** (excess energy, choline/methionine deficiency), **prolapse** (over-fat, early maturity). In heat stress add **sodium bicarbonate** and raise nutrient density because intake falls.

DUCKS, QUAILS & TURKEYS

BIRD	STARTER CP	DISTINCTIVE REQUIREMENT
Duck	20–22%	Markedly higher niacin (~55 mg/kg) — deficiency causes bowled legs and leg weakness . Prefers wet mash and needs water to swallow. Better able to use fibrous and unconventional feeds; excellent scavenger in paddy fields
Japanese quail	27%	The highest protein requirement of any domestic bird ; very small particle size; matures at 6 weeks and lays at 6–7 weeks
Turkey	28%	Very high protein and lysine early; prone to starve-out (poult must be taught to eat and drink); needs high niacin and riboflavin

Unconventional feed ingredients for poultry & swine — name and limit them: de-oiled rice bran, maize gluten, sunflower and safflower cake, **sal seed meal** (tannins), **mahua cake** (saponins), **salseed / karanj cake** (bitter principles), **silkworm pupae meal** (high protein, rancidity risk), **azolla**, spirulina, single-cell protein, **slaughterhouse and poultry by-product meal**, dried bakery waste, molasses, tapioca/cassava (HCN — must be sun-dried), and fruit/vegetable processing waste. The three limits that must always be respected: **fibre content, anti-nutritional factors, and consistency of supply**.

POULTRY DEFICIENCY DISORDERS — THE PATHOGNOMONIC LIST

SIGN	DEFICIENCY
Perosis / slipped tendon	Manganese (also choline, biotin, folic acid, niacin)
Curled-toe paralysis	Riboflavin (B ₂)
Encephalomalacia — "crazy chick"	Vitamin E
Exudative diathesis	Vitamin E + Selenium
Fatty Liver & Kidney Syndrome	Biotin
Rickets / soft beak & bones	Vitamin D ₃ , Ca or P
Cage layer fatigue	Ca, P, Vitamin D ₃
Nutritional roup, xerophthalmia	Vitamin A
"Spectacle eye", dermatitis of feet	Pantothenic acid (B ₅)
Bowed legs in ducklings	Niacin
Haemorrhage, poor clotting	Vitamin K
Poor feathering, megaloblastic anaemia	Folic acid

Never confuse these three: Curled toe = B₂ · Perosis = Mn · Crazy chick (encephalomalacia) = Vitamin E.

RABBIT & LABORATORY ANIMALS

RABBIT — hind-gut fermenter with a unique trick. Diet: **CP 16–18%, crude fibre 12–16%** (a minimum of 12% is essential — low fibre causes enteritis and mucoid enteropathy), ME ~2500 kcal/kg, Ca 0.5–1.0%. Fed as pellets plus green fodder.

Caecotrophy (coprophagy) — always asked. The rabbit produces **two kinds of faeces**: hard, dry pellets voided during the day, and soft, mucus-coated **caecotrophs (night faeces)** which it takes directly from the anus and swallows whole. These caecotrophs are rich in **microbial protein and B vitamins** synthesised in the caecum, and re-ingesting them recovers nutrients that would otherwise be lost past the site of absorption. Preventing caecotrophy (e.g. by a collar) causes growth failure — proof that it is nutritionally essential, not a vice.

LAB ANIMAL	CP	DISTINCTIVE REQUIREMENT
Rat	20–24%	Omnivore; practises coprophagy for B vitamins; the classic model for PER and BV assays
Mouse	18–20%	High metabolic rate ⇒ high energy density; needs frequent access to feed & water
Guinea pig	18–20%	Requires dietary Vitamin C (10–30 mg/day) — cannot synthesise it ⇒ scurvy; also needs high fibre (~15%) and, being a strict herbivore, is very sensitive to antibiotic-induced enterotoxaemia
Hamster / gerbil	16–20%	Prone to obesity; hoards feed

General laboratory-animal principles: diets must be **chemically defined, batch-consistent, sterilisable (autoclaving destroys vitamins ⇒ over-fortify), and free from contaminants**, because the animal's nutrition is an experimental variable. Provide *ad libitum* clean water and a fixed photoperiod.

DOG & CAT

DOG — an omnivorous carnivore. Adult maintenance **CP 18–22%**, fat 8–15%; puppy/lactating bitch **CP 25–30%**. Energy: **ME ≈ 130 kcal × W^{0.75}** per day for an adult; puppies and lactating bitches need 2–4×. **Dogs have no dietary carbohydrate requirement** (glucose can be made from protein and glycerol) but tolerate starch well. Requires **arginine**; some breeds need **taurine** (deficiency ⇒ dilated cardiomyopathy). Feed adults once or twice daily; large breeds need **controlled growth and correct Ca:P** — over-supplementing calcium causes developmental orthopaedic disease.

THE CAT — obligate carnivore. The seven peculiarities (the single most-asked question in Unit 4):

① **High protein requirement** — adult **26–30%+**, kitten **30–35%**; hepatic transaminase and urea-cycle enzymes are permanently high and **cannot be down-regulated**, so protein is catabolised for energy even during fasting.

② **Taurine is dietary-essential** — cats have low cysteine sulphinic acid decarboxylase and conjugate bile acids only with taurine. Deficiency ⇒ **feline central retinal degeneration (blindness), dilated cardiomyopathy**, and reproductive failure.

③ **Arginine is critically essential** — a **single arginine-free meal** can cause hyperammonaemia, tremors and death, because the cat cannot synthesise ornithine.

④ **Cannot convert β-carotene to Vitamin A** (no intestinal dioxygenase) ⇒ needs **preformed retinol** from animal tissue.

⑤ **Cannot convert tryptophan to niacin** ⇒ requires about **4× the dog's dietary niacin**.

⑥ **Arachidonic acid is essential** — the cat lacks Δ6-desaturase and cannot make it from linoleic acid.

⑦ **No sweet taste receptor** and limited glucokinase ⇒ indifferent to sugars and poorly adapted to high-carbohydrate diets.

Feline clinical links: **hepatic lipidosis** in an obese cat that stops eating; **feline lower urinary tract disease / struvite urolithiasis** — controlled by urinary acidification, restricted magnesium and, above all, **increased water intake (wet food)**. Cats are nibblers — offer several small meals.

FEEDING WILD & CAPTIVE ZOO ANIMALS

The four governing problems in captivity: ① the natural diet is unknown or unobtainable, so we feed an approximation; ② **activity is greatly reduced** ⇒ obesity is the commonest nutritional disorder in a zoo; ③ animals are exposed to unnatural feeds and to visitor feeding; ④ deficiencies appear slowly and are often diagnosed only at post-mortem.

GROUP	FEEDING PRACTICE & THE CLASSIC PITFALL
Large carnivores (tiger, lion, leopard)	8–10 kg meat/day for a tiger, with bone — muscle meat alone is severely Ca-deficient (Ca:P about 1:20) and causes nutritional secondary hyperparathyroidism / rickets . Give one fasting day per week to mimic natural feeding and prevent obesity and gut problems. Supplement Vitamin A, taurine and calcium
Fish-eating species (otter, penguin, seal, piscivorous birds)	Raw fish contains thiaminase ⇒ thiamine deficiency; supplement thiamine and Vitamin E (fish oil oxidation causes steatitis / yellow fat disease)
Herbivores (elephant, deer, antelope, rhino)	Elephant: 150–200 kg green fodder + 10–20 kg concentrate daily. Provide browse and long fibre for gut health and to prevent stereotypy; mineral licks
Primates	Require dietary Vitamin C like the guinea pig; provide fruit, vegetables and a balanced primate pellet; avoid feeding only fruit (low protein, high sugar ⇒ obesity and diabetes)
Reptiles	Metabolic bone disease is the classic captive disorder — from low Ca, inverted Ca:P and, crucially, absent UV-B light which reptiles need to synthesise Vitamin D ₃ . Provide UV-B, calcium dusting and gut-loaded insects
Psittacine & other birds	An all-seed diet is deficient in Ca, Vitamin A, lysine and methionine ⇒ feather picking, hypovitaminosis A, poor bone. Feed formulated pellets plus greens

General zoo principles to write: mimic the **natural diet, feeding frequency and feeding behaviour**; use **environmental/feeding enrichment** (scatter feeding, whole carcass, puzzle feeders) because feeding behaviour is a welfare need, not just nutrition; maintain **strict hygiene and feed quality**; monitor **body weight and condition** routinely; adjust for season, breeding and hand-rearing; and keep species-wise diet charts with periodic veterinary review.

THE ONE TABLE THAT ANSWERS MOST UNIT 4 QUESTIONS				
SPECIES	DIGESTIVE TYPE	TYPICAL CP	ENERGY / INTAKE NOTE	THE ONE THING THAT IS UNIQUE
Pig	Simple monogastric omnivore	14–22%	ME 3000–3300 kcal/kg; FCR 2.5–3.5	Lysine first limiting; iron injection on day 3; parakeratosis from Zn/Ca imbalance
Horse	Hind-gut fermenter (caecum & colon)	10–18%	DM 1.5–3% BW; roughage minimum 1% BW	Small stomach, no gall bladder, cannot vomit; monensin is fatal
Broiler	Avian, gizzard, rapid passage	20–23%	ME 2800–3200; FCR 1.5–1.7	Methionine first limiting; needs Vitamin D₃ , not D ₂
Layer	Avian	18%	110–120 g feed/day	Ca 3.5–4% , coarse particulate limestone fed in the evening
Duck	Avian, waterfowl	20–22%	Wet mash; good scavenger	High niacin (~55 mg/kg)
Quail	Avian	27%	Tiny particle size	Highest CP requirement of domestic birds
Turkey	Avian	28%	Poultz prone to starve-out	Very high early protein & lysine
Rabbit	Hind-gut fermenter	16–18%	Needs CF 12–16%	Caecotrophy — eats its soft night faeces
Guinea pig	Hind-gut fermenter	18–20%	High fibre	Dietary Vitamin C essential
Rat / Mouse	Omnivore	18–24%	Coprophagy supplies B vitamins	Reference species for PER & BV assays
Dog	Omnivorous carnivore	18–22% adult	ME ≈ 130 × W ^{0.75} kcal/day	No dietary CHO requirement; arginine essential
Cat	Obligate carnivore	26–35%	Nibbler — several small meals	Taurine, arachidonic acid, preformed Vitamin A, high niacin, high protein

LAST-10-MINUTES DRILL — UNIT 4

First limiting AA in pigs — **Lysine**
First limiting AA in poultry — **Methionine**
Ideal protein reference AA — **Lysine = 100**
Piglet iron dose — **150–200 mg iron-dextran IM, day 3**
Creep feed start, piglet — **7–10 days**
Pig weaning age — **4–8 weeks**
Pregnant sow feeding — **Restricted, ~2 kg/day, CP 14%**
Lactating sow feeding — **2 kg + 0.25 kg per piglet**
Pig FCR — **2.5–3.5**
Parakeratosis in pigs — **Zn deficiency (excess Ca)**
Goose-stepping — **Pantothenic acid**
Mulberry heart — **Vitamin E / Se**
Horse stomach capacity — **8–15 L (~10% of GIT)**
Organ horses lack — **Gall bladder**
Minimum roughage for horse — **1% of body weight**
Monday morning disease — **Azoturia / tying-up**
Big head in horses — **Oxalate ⇒ osteodystrophia fibrosa**
Ionophore fatal to horses — **Monensin**
Broiler starter CP & ME — **23%, 2800–3000 kcal/kg**
Broiler finisher CP — **20%**
Broiler FCR & market weight — **1.5–1.7; 1.8–2.2 kg at 5–6 wk**
Layer mash CP & Ca — **18% CP, 3.5–4% Ca**
Layer daily feed intake — **110–120 g**
Calcium in one egg shell — **~2 g**
Why coarse limestone — **Slow night release for shell calcification**
Calorie:protein, broiler starter — **~130–140**
Cattle feed BIS Type I — **CP 22%, TDN 70%**
Cattle feed BIS Type II — **CP 20%, TDN 65%**
Max AIA in cattle feed — **4%**
Duck distinctive vitamin — **Niacin ~55 mg/kg**
Highest CP among birds — **Quail 27% / turkey 28%**
Rabbit minimum crude fibre — **12%**
Caecotrophy — **Rabbit eating soft night faeces**
Guinea pig special need — **Dietary Vitamin C**
Dog maintenance energy — **~130 kcal × W^{0.75}**
Nutrient dogs don't need in diet — **Carbohydrate**
Cat taurine deficiency — **Retinal degeneration + DCM**
One arginine-free meal in cat — **Hyperammonaemia, possibly fatal**
Vitamin cats need preformed — **Vitamin A (no carotene conversion)**
Cat & tryptophan — **Cannot make niacin ⇒ needs 4× the dog's**
Fatty acid essential for cats — **Arachidonic acid**
Adult cat dietary CP — **26–30%+**
Obese anorectic cat disease — **Hepatic lipidosis**
Tiger daily meat & fasting — **8–10 kg with bone; 1 fast day/week**
All-muscle-meat diet causes — **Nutritional secondary hyperparathyroidism**
Raw fish diet risk — **Thiaminase ⇒ thiamine deficiency**
Reptile captive disorder — **Metabolic bone disease (no UV-B/D₃)**
Primates special need — **Dietary Vitamin C**
Elephant green fodder/day — **150–200 kg**
Commonest zoo nutritional disorder — **Obesity**
Enzyme freeing phytate P — **Phytase**